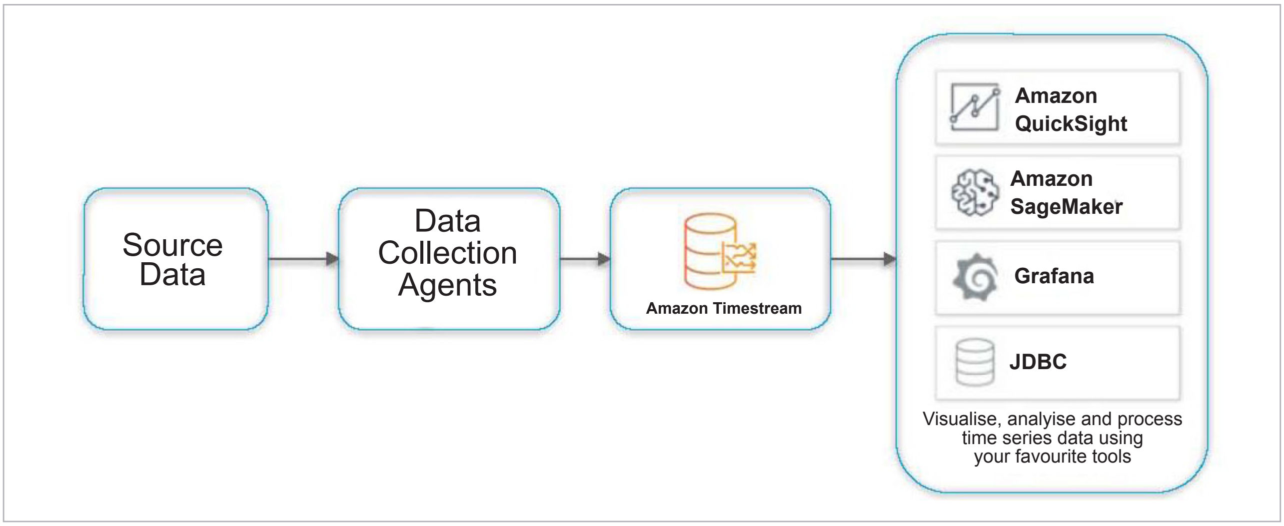

Figure 2: Time series analytics engine on AWS Cloud

for cheaply evicting large sets of data and constantly summarising that data at scale. With a time series database, this functionality is provided out-of-the-box.

Since time series data comes in time order and is typically collected in real-time, time series databases are immutable and append-only to accommodate extremely high volumes of data. This append-only property distinguishes time series databases from relational databases, which are optimised for transactions but only accommodate lower ingest volumes. In general, depending on their particular use case, NoSQL databases will trade off the ACID principles for a BASE model (whose principles are basic availability, soft state and eventual consistency). For example, one individual point in a time series is fairly useless in isolation, and the important thing is the trend in total.

Alerts based on time series data analysis for site reliability

Time series data models are very common in site reliability engineering. Time series analysis is used to monitor system health, performance, anomaly detection, security threat detection, inventory forecasting, etc. Figure 1 shows a typical alerting mechanism based on analysing time series data collected from different components.

Modern data centres are complex systems with a variety of operations and analytics taking place around the clock. Multiple teams need access at the same time, which requires coordination. In order to optimise resource use and manage workloads, systems administrators monitor a huge number of parameters with frequent measurements for a fine-grained view. For example, data on CPU usage, memory residency, IO activity, levels of disk storage, and many other parameters are all useful to collect as time series.

Once these data sets are recorded as time series, data centre operations teams can reconstruct the circumstances that lead to outages, plan upgrades by looking at trends, or even detect many kinds of security intrusions by noticing changes in the volume and patterns of data transfer between servers and the outside world.

Open source TSDBs

Time series databases are the fastest growing segment in the database industry. There are many commercial and open source time series databases available. A few well-known open source time series databases are listed below:

- **InfluxDB** is one of the most popular time series open source databases, and is written in Go. It has been designed to provide a highly scalable data ingestion and storage engine. It is very efficient at collecting, storing, querying, visualising, and taking action on streams of time series data, events, and metrics in real-time. It uses InfluxQL, which is very similar to a structured query language, for interacting with data.
- **Prometheus** is an open source monitoring solution used to understand insights from metrics data and send the necessary alerts. It has a local on-disk time-series database that stores data in a custom format on disk. It provides a functional query language called PromQL.
- **TimescaleDB** is an open source relational database that makes SQL scalable for time series data. This database is built on PostgreSQL.
- **Graphite** is an all-in-one solution for storing and efficiently visualising real-time time series data. Graphite can store time series data and render graphs on demand. To collect data, we can use tools such as collectd, Ganglia, Sensu, telegraf, etc.